Enhancing Perturbation Modeling in Single-Cell Data through Advanced Deep Learning Approaches

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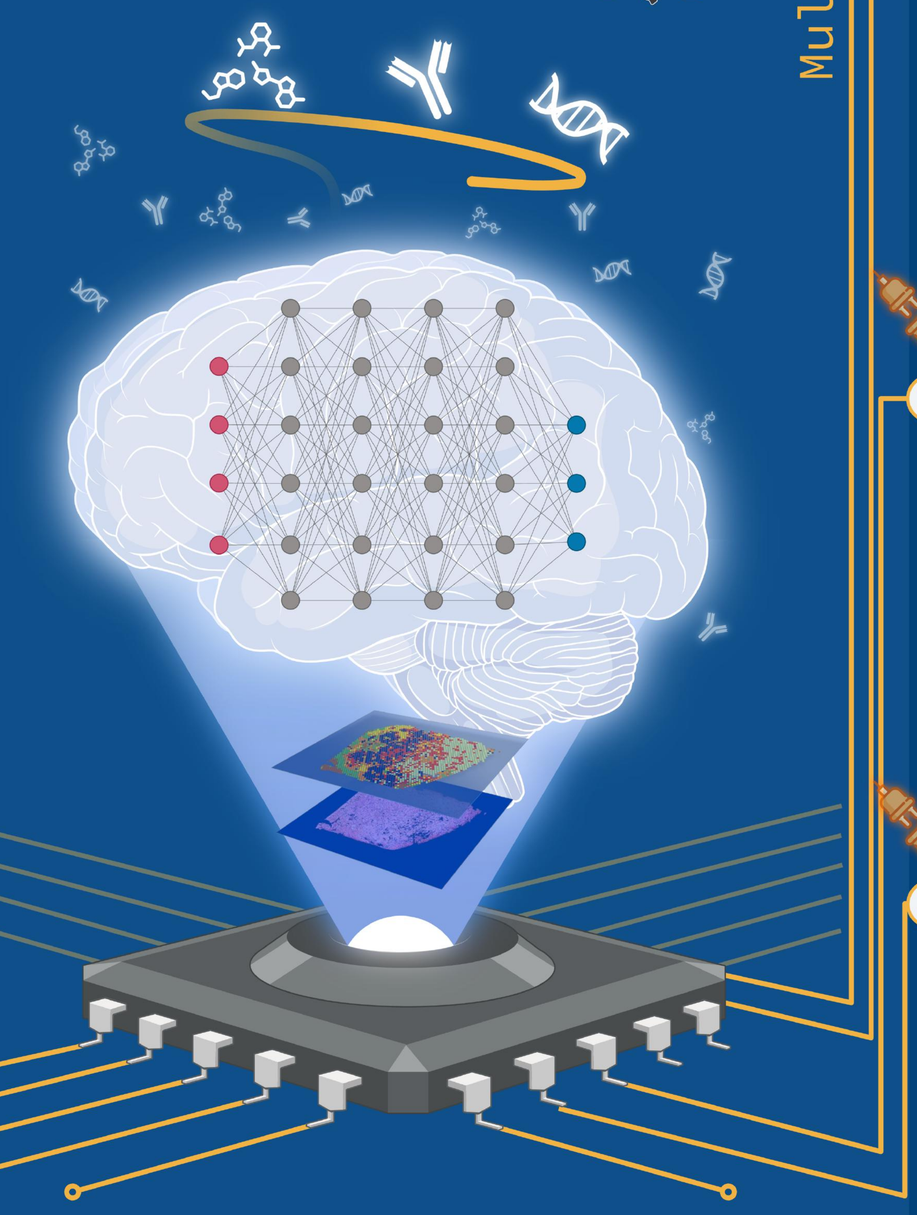
AI in single-cell spatial omics for Systems Medicine

Artificial intelligence (AI) can perform pattern recognition across high-dimensional single-cell and spatial data, dissecting the innate cellular heterogeneity of various tissues and organs, in health and disease.

Some of the tasks that AI addresses in single-cell biology are:

1. Joint group identification
2. Cross-modal prediction
3. Perturbation modelling
4. Pathway enrichment

These tasks are usually pursued in isolation, offering partial glimpses of perturbed cell physiology!



Multi-Task Learning

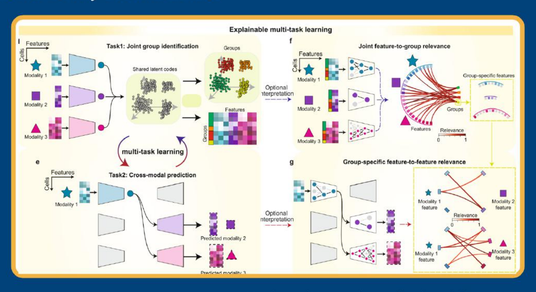
The Mongoose framework

Multi-Objective Network Generator Of Optimized Single-cell Experiments

Explore the impact of Multi-Task Learning (addressing multiple tasks during model training) in perturbation modelling

1 Joint group identification

AI can analyse omic modalities from single cell/spatial data and discern distinct cell groups and cell states. Combining multiple omic modalities enhances the ability of AI to detect more fine-grain populations.



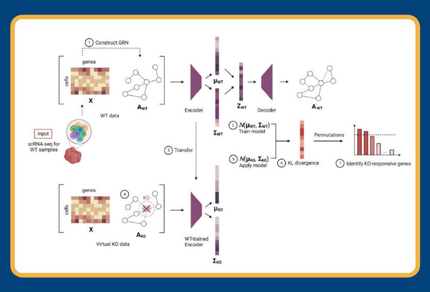
UnitedNet (explainable MTL on single-cell/spatial omics)

2 Cross-modal prediction

AI on single-cell multi-modal data provides the opportunity to discover cell-type-specific relationships between gene expression and other modalities, helping to uncover the regulatory mechanisms underlying the biological condition of interest.

3 Perturbation modelling

AI can delineate cellular perturbations (drugs, genetic knock-outs/ins) and then predict the effect of these perturbations on cells that never actually experienced them (extrapolation to unseen events).
Also, AI can unmask true perturbation effects from confounding sources of variation (technical-biological).
Moreover, AI can predict digital perturbations without prior knowledge and can approximate synergies or antagonistic effects.



GenKI (Gene knockout inference with VGAE learning single-cell GRNs)

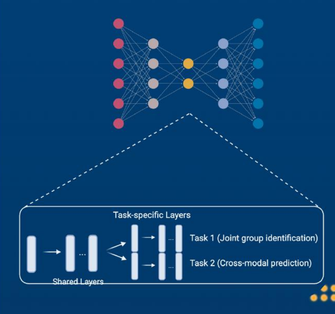
4 Pathway enrichment

AI approaches can illuminate the underlying biology of the single-cell and spatial datasets, highlighting differentially expressed genes or proteins and providing insights for significantly altered signaling pathways.

reactome, GENEONTOLOGY, KEGG, WikiPathways, STRING

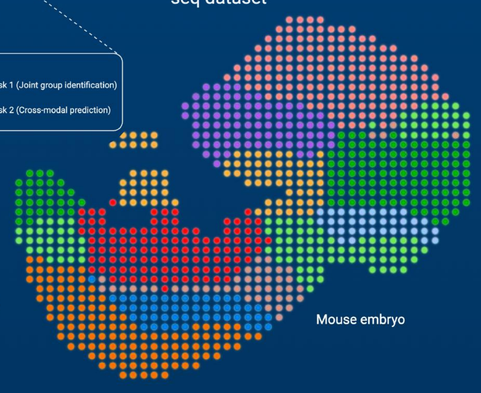
STRINGdb database

Deploying Mongoose in healthy DBiT-seq for kidney perturbations



Tasks 1-2. Through UnitedNet, Mongoose performs Joint Group Indentification and Cross-Modal Prediction on the multi-omic DBiT-seq dataset

Three modalities:
- RNA
- protein
- "niche modality" (spatially-aware gene expression)

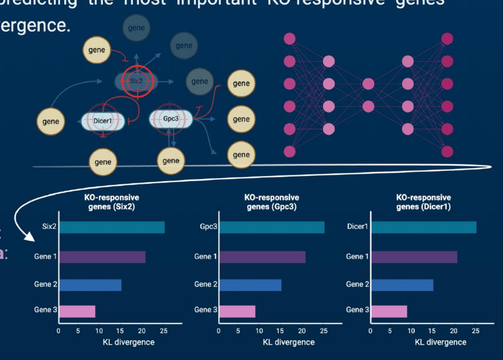


SHAP values for cross-modal predictions for Cluster 1-12

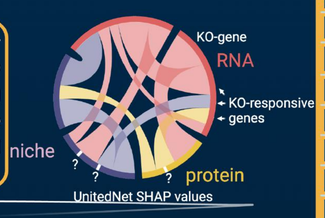
Bridging over to perturbation modelling (digital KO)

Task 3. Through GenKI, Mongoose reconstructs the DBiT-seq Gene Regulatory Network (GRN) and performs digital-KOs for selected targets (one at at time), predicting the most important KO-responsive genes based on the KL divergence.

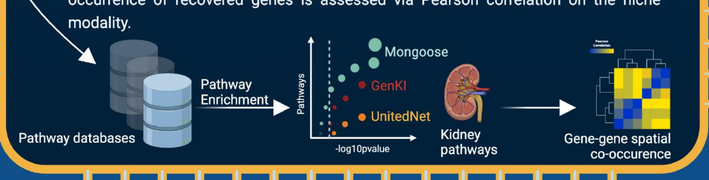
Digital KOs: **targeting genes related to normal kidney development and nephroblastoma**:
- Six2
- Dicer1
- Gpc3



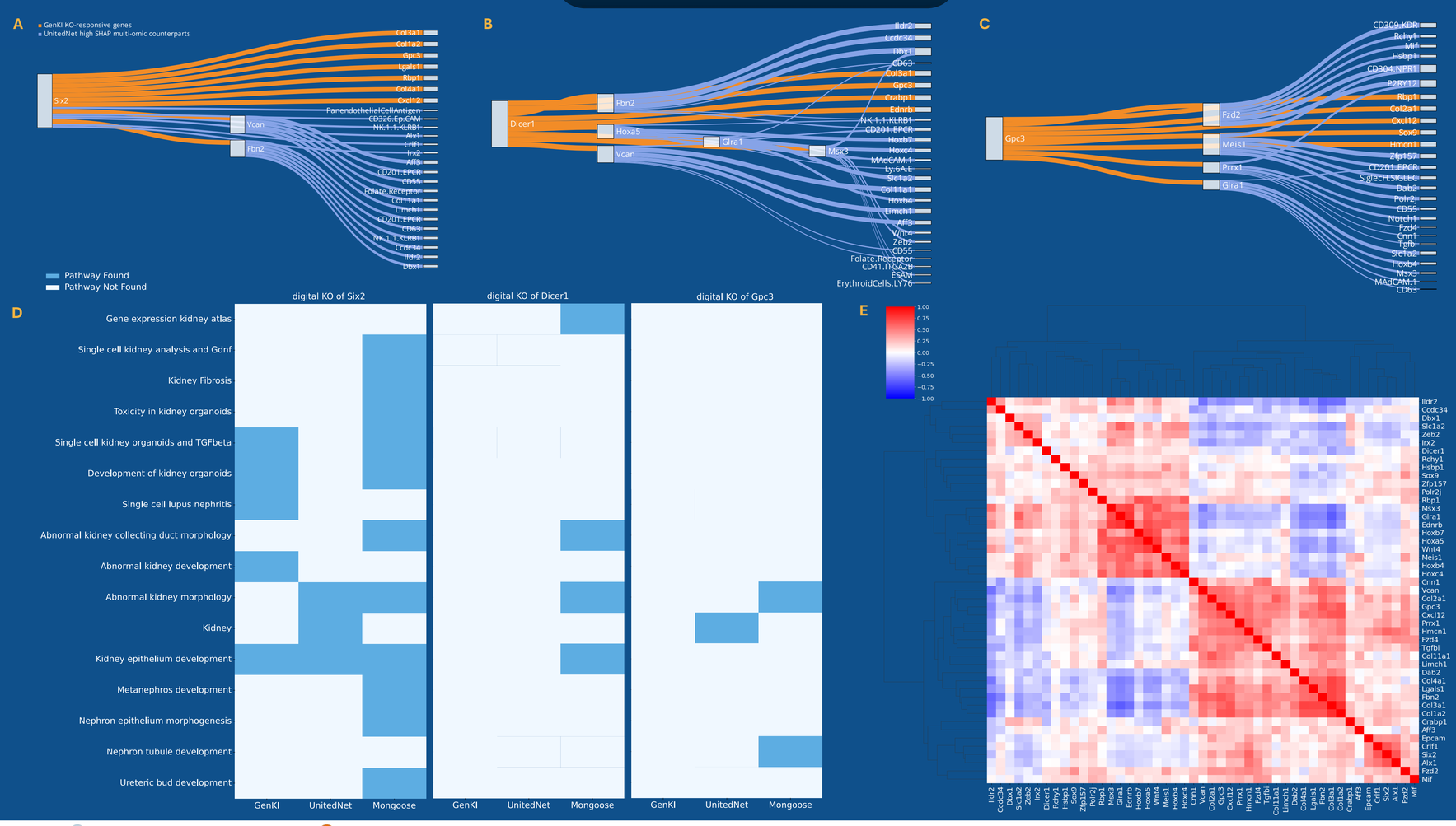
Task 4. Mongoose maps the GenKI KO-gene target and KO-responsive genes to other modalities (protein, niche) using the UnitedNet SHAP values and the provides through STRINGdb biological insights.



Perturbed molecules from RNA, protein, and niche modalities predicted by Mongoose undergo STRINGdb enrichment to identify kidney-related pathways. Comparisons against GenKI and UnitedNet baselines then ensue. Lastly, spatial co-occurrence of recovered genes is assessed via Pearson correlation on the niche modality.



Perturbational predictions



Highlights

1. **(A-C)**: Alluvial plots showing the mappings of digital KO targets Six2, Dicer1 and Gpc3 to several KO-responsive genes through GenKI (orange). KO-targets and KO-responsive genes are further mapped to multi-omic counterparts through high UnitedNet SHAP values (pink) (genes from niche modality and proteins). In this way, the Mongoose framework extends the "signal" of the digital perturbation across multiple omic modalities.

2. **(D)**: Heatmaps showing the number of kidney related pathways retrieved by baseline models (GenKI, UnitedNet) and by the Mongoose approach which outperforms them in every scenario.

3. **(E)**: Hierarchical clustering heatmap depicting Pearson correlation co-efficients for gene-gene spatial co-occurence in the DBiT-seq dataset across all 3 scenarios. The prioritised markers that Mongoose is predicting in kidney perturbations exhibit spatial patterning that should be further researched.

Future Directions

4. Future implementations of Mongoose could be extended to multi-omic Perturb-seq and sciplex single-cell datasets which emerged from single-cell CRISPR and drug high-throughput screening experiments respectively.

5. Mongoose could also other spatial technologies (e.g. Visium HD) across a variety of disease settings to uncover perturbationally-relevant insights.

6. Mongoose could catalyze the creation of advanced foundational models that combine transformers and MTL to dovetail perturbation tasks with other single-cell/spatial tasks (e.g. niche identification, cell-state determination) offering more powerful predictions, closer to the biological ground-truth.

For more info

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